

Sound Processes

Composing with Algorithms
<http://www.bjarni-gunnarsson.net>

Exercises - Synthesis

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1. Extend example 1 by making the rate of the impulse train change in time as well as the amount of resonance.
2. Extend example 2 by giving a choice of UGens instead of only using sines.
3. Extend example 3 by using different methods to give random frequency values then the exponential one. Additionally experiment with Allpass and Delay times.
4. Extend example 6 by making the resonance appear gradually and with slight modulation that form some sort of relationship to the excitation source.
5. Extend example 8 by adding multiple oscillators.

Exercises - Synthesis

6. Implement a sound generator for synthetic impulse trains that can give a sense of rhythmic activity.
7. Implement a looping device where the loop length changes in time and a pattern that executes the change.
8. Create a network of synthesis UGens where different oscillators, with different filters are used to create the sound output.
9. Integrate some of the processing ideas from last time with a synthesis SynthDef to form a chain of UGens.

Exercises - Processing

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1. Extend example 1 by adding further distortion modules and ways of dividing the frequency spectrum.
2. Extend example number 3 by adding more pitch shifters that contain some sort of ratio or relation.
3. Extend example number 5 by substitution the FreqShift modulation with amplitude modulation, preferably more than one source.
4. Extend example number 7 by using the pitch and amplitude tracking to control noise sources instead of oscillators.
5. Extend example number 8 so that the amount of convolution varies either by tracking the incoming signal or using modulation.

Exercises - Processing

6. Implement a processing unit that will decrease the spectral fill factor of an incoming signal (perhaps filtering) while adding to the tail of the sound so its temporal duration is extended.
7. Implement a processing unit that will make an incoming signal more rich (perhaps by using distortion) while limiting the amplitude of the output sound so that it will appear with less temporal weight (perhaps with gaps).
8. Implement a processing unit that takes two sampled sound sources as input and uses one to modulate the other.
9. Implement a processing unit that uses feedback techniques to give tail to granular incoming sound.

